

Modal Compute Guide

CS224R (Spring 2026)

Adapted from CS336 Spring 2026 Modal Compute Guide

1 Purpose and scope

This document is a practical introduction to [Modal](#) for running Python (especially GPU) workloads.

What this guide covers: authentication, defining images and apps, running functions remotely, volumes, parallelism (`map` / `starmap`), GPUs, secrets, shells, and finding logs in the Modal UI.

2 Quickstart

1. Create a Modal account (e.g. sign in with Google using your Stanford email, unless staff say otherwise).
2. Redeem the course compute code from the email from the teaching team with subject **Your CS224R Modal Compute Credits** via <https://modal.com/credits> (if you have not received credits yet please fill out <https://forms.gle/gYkTgAqkEvd9Ybcj7>).
3. Install the CLI in the project where you work:

```
uv add modal
# or: pip install modal
```

4. Authenticate:

```
uv run modal setup
# or: modal setup
```

5. Run a small script locally, then the same file with `modal run` once you have wrapped work in `@app.function` / `@app.local_entrypoint` (see below).

3 Security and good practice

- Do not commit API keys, passwords, or private dataset paths into `git`. Prefer [Modal secrets](#) for tokens (Section ??).
- Your Modal account and runs are yours; still follow Stanford's honor code and each assignment's collaboration rules for code and writeups.

4 Turning a local script into a Modal job

A practical recipe:

1. Keep underlying Python logic unchanged when possible.
2. Define a `modal.App` and an `Image` with the packages you need (PyTorch, Gymnasium, simulators, etc.).
3. Decorate expensive work with `@app.function(...)`. Set CPU or GPU, volumes, and timeouts as needed.
4. Add `@app.local_entrypoint()`. Start it from your terminal with `modal run`, using the script path your handout gives.
5. From that entrypoint, call `.remote(...)`, `.map(...)`, or `.starmap(...)` so heavy work runs remotely, not inline on your laptop.

4.1 Fill-in-the-blanks template

Most of the assignments in this class look the same from Modal's point of view: there is a training script you already run locally, and you want the whole thing to happen on a remote machine—maybe with a GPU—and you want the logs to survive after the job ends. The template below handles all of that. Every line marked `# <-- CHANGE` is something you swap out for your own homework.

```
import modal
import subprocess, shutil

app = modal.App("cs224r-hwN") # <-- CHANGE

# 1) Image -- your pip (and optionally apt) dependencies.
# Check the requirements.txt or conda_env.yml in your hw repo.
image = (
    modal.Image.debian_slim(python_version="3.10")
    # If the assignment uses MuJoCo or offscreen rendering, uncomment:
    # .apt_install("libgl1-mesa-glx", "libosmesa6-dev", "patchelf")
    # .env({"MUJOCO_GL": "egl"})
    .pip_install( # <-- CHANGE: whatever is in your conda_env.yml (or
                  # alternatively you can make a requirements.txt and just list that)
                  "torch", "numpy", "gym==0.25.2", "tensorboard",
    )
)

# 2) Volume -- this allows checkpoints to persist after container shutdown
volume = modal.Volume.from_name("cs224r-results", create_if_missing=True)

# 3) Mount -- ships your local homework folder to the container.
code = modal.Mount.from_local_dir(
    ".", # <-- CHANGE: path to your hw folder on your laptop
    remote_path="/root/hw",
)

@app.function(
```

```

        image=image,
        mounts=[code],
        volumes={"/output": volume},
        gpu="T4",          # <-- CHANGE: to another GPU type if you want
        timeout=3600,      # <-- CHANGE: how long (seconds) before Modal kills
                           the run
    )
def train():
    import os
    os.chdir("/root/hw")

    # <-- CHANGE: whatever command you would run on your own machine
    subprocess.run(
        ["python", "your_script.py", "--your_flag", "value"],
        check=True,
    )

    # Copy the output directory your script wrote to the persistent volume
    # . This is the directory that will persist after the container shuts
    # down.
    shutil.copytree("data", "/output/hwN_run", dirs_exist_ok=True) # <--
    CHANGE
    volume.commit()

@app.local_entrypoint()
def main():
    train.remote()

```

Then just run `modal run your_modal_file.py`. Your code goes to the container through the mount, training runs there, and the results land on the volume. You can look at them afterwards with `modal shell` or from another job.

4.2 Volumes and persistence

The local disks of workers running Modal code are ephemeral. After a job finishes, all the data you save is deleted. To persist files, you use Modal Volumes . This is where you put your datasets, preprocessed data, checkpoints, and evaluation outputs.

```

import modal
from pathlib import Path

app = modal.App("volume-demo")
data_vol = modal.Volume.from_name("my-volume", create_if_missing=True)
image = modal.Image.debian_slim()

@app.function(image=image, volumes={"/data": data_vol}) # mount data_vol
                at /data; files there persist for later jobs
def write_data() -> None:
    Path("/data/data.txt").write_text("this will be persisted")
    data_vol.commit()

```

We highly suggest you read the official volume guide for your Modal version.

4.3 Runnning Jobs in Parallel

If the same function should run independently on many inputs, use `.map` .

```
@app.local_entrypoint()
def main():
    for path in download_or_train.map(list_of_jobs):
        print(path)
```

This is cleaner than manually looping over `.remote(...)` calls, and it makes the "one function, many inputs" pattern obvious.

Large sweeps can spend credits quickly and may queue during busy periods. Use `max_containers` to cap how many workers run at once:

```
@app.function(image=image, volumes={"/data": data_vol}, max_containers=3)
def run_experiment(seed: int):
    ...
    return result

@app.local_entrypoint()
def sweep():
    for r in run_experiment.map(range(12)):
        print(r)
```

5 Example: toy behavioral cloning with a placeholder environment

Heere is an example of how to code you've already written to run on Modal.

Local script (single run):

```
import gymnasium as gym
import numpy as np
from sklearn.tree import DecisionTreeClassifier

def collect_expert_data(env_id: str, n_episodes: int = 50):
    env = gym.make(env_id)
    observations, actions = [], []
    for _ in range(n_episodes):
        obs, _ = env.reset()
        done = False
        while not done:
            action = 1 if obs[2] > 0 else 0
            observations.append(obs)
            actions.append(action)
            obs, _, terminated, truncated, _ = env.step(action)
            done = terminated or truncated
    return np.array(observations), np.array(actions)

def main():
    env_id = "YOUR_ENV"
    obs, actions = collect_expert_data(env_id, n_episodes=50)
```

```

model = DecisionTreeClassifier(max_depth=5)
model.fit(obs, actions)

env = gym.make(env_id)
obs, _ = env.reset()
total_reward = 0
for _ in range(500):
    action = int(model.predict([obs])[0])
    obs, reward, terminated, truncated, _ = env.step(action)
    total_reward += reward
    if terminated or truncated:
        break
print("Total reward:", total_reward)

if __name__ == "__main__":
    main()

```

Modal version: parallel hyperparameter grid. Each remote call is self-contained (collect, train, evaluate), so you do not ship large tensors from your laptop. `.starmap` dispatches many jobs subject to your limits and `max_containers`. A small [Modal Volume](#) stores one JSON file per config under `/data/runs/` so you can inspect past sweeps from a modal shell or a later job without re-running.

```

import json
import modal
import numpy as np
from pathlib import Path

app = modal.App("bc-sweep-demo") # logical grouping for Modal dashboard

bc_results = modal.Volume.from_name(
    "bc-sweep-demo-results", create_if_missing=True
) # durable disk shared across runs (not the ephemeral container FS)

image = (
    modal.Image.debian_slim()
    .pip_install("gymnasium", "scikit-learn", "numpy")
) # dependencies installed on the remote worker image

# Body runs on Modal; /data is the volume mount; max_containers caps
# concurrent workers
@app.function(image=image, volumes={"/data": bc_results}, max_containers=3)
def train_and_eval(n_episodes: int, max_depth: int) -> dict:
    import gymnasium as gym # import here so the worker image resolves them
    from sklearn.tree import DecisionTreeClassifier

    env_id = "YOUR_ENV"

    def collect_expert_data(n_episodes):
        env = gym.make(env_id)
        observations, actions = [], []
        for _ in range(n_episodes):

```

```

        obs, _ = env.reset()
        done = False
        while not done:
            action = 1 if obs[2] > 0 else 0
            observations.append(obs)
            actions.append(action)
            obs, _, terminated, truncated, _ = env.step(action)
            done = terminated or truncated
        return np.array(observations), np.array(actions)

obs, actions = collect_expert_data(n_episodes)
model = DecisionTreeClassifier(max_depth=max_depth)
model.fit(obs, actions)

env = gym.make(env_id)
obs, _ = env.reset()
total_reward = 0
for _ in range(500):
    action = int(model.predict([obs])[0])
    obs, reward, terminated, truncated, _ = env.step(action)
    total_reward += reward
    if terminated or truncated:
        break

result = {
    "n_episodes": n_episodes,
    "max_depth": max_depth,
    "reward": float(total_reward),
}

run_dir = Path("/data/runs") # lives on the mounted volume, not local
                             # disk
run_dir.mkdir(parents=True, exist_ok=True)
out_path = run_dir / f"n{n_episodes}_d{max_depth}.json"
out_path.write_text(json.dumps(result)) # one file per hyperparameter
                                         # pair
bc_results.commit() # flush volume writes so the next job sees them

return result

@app.local_entrypoint() # 'modal run this_file.py' starts here on your
                        # laptop
def main():
    configs = [
        {"n_episodes": n, "max_depth": d}
        for n in [10, 50, 200]
        for d in [2, 5, 10]
    ]
    results = list(train_and_eval.starmap(configs)) # parallel remote
                                                    # calls; cap via max_containers
    best = max(results, key=lambda r: r["reward"])
    print("Best config:", best)

```

For heavier models, reuse the same volume pattern for checkpoints (serialized with `pickle`, `joblib`, or `PyTorch`). Pick a volume name that is unique to you if you share a Modal org with others.

6 GPU jobs

Request a GPU on the decorator for remote runs. Valid `gpu=` strings depend on Modal's current offerings and your account; check [Modal's GPU documentation](#).

```
import torch

def do_work():
    device = "cuda" if torch.cuda.is_available() else "cpu"
    x = torch.randn(1024, 1024, device=device)
    return x.sum().item()

@app.function(image=gpu_image, gpu="YOUR_GPU_TYPE") # example gpu="A100"
def remote_train():
    return do_work()
```

The `gpu="YOUR_GPU_TYPE"` line only affects remote runs, e.g. `run.with_gpu.remote()`.

`run.with_gpu.local()` runs in your local Python process and uses whatever CPU or GPU your laptop or workstation already has.

7 Interactive shell

[Modal shell](#) starts a shell with the same image (and optional volumes) as a given function—handy for debugging imports or inspecting mount paths:

```
modal shell path/to/your_script.py::your_function
```

8 Secrets

Do not print secret values. Define secrets in the CLI and attach them to functions:

```
modal secret create wandb WANDB_API_KEY=your_token_here
```

```
import os
import modal

wandb_secret = modal.Secret.from_name("wandb")

@app.function(image=my_image, secrets=[wandb_secret])
def log_run():
    _ = os.environ["WANDB_API_KEY"]
    ...
```

9 Dashboard and logs

Open modal.com, sign in, and browse your apps and run history (logs, failures, timing). Your workspace is individual; use whatever filters the UI provides to find recent jobs.

10 Summary

- Redeem course credits, run `modal setup`, then define an `App`, `Image`, and `@app.function` entry-points.
- Use volumes for checkpoints and data; treat default worker filesystems as temporary.
- Use `map` / `starmap` for parallel jobs; `max_containers` helps control cost and concurrency.
- Attach `gpu=` for remote GPU; use secrets for API keys; use `modal shell` to debug the remote environment.
- This guide is about Modal mechanics—assignment-specific choices stay in your own code.